

9

NUMBER AND ALGEBRA

EQUATIONS

Equations are extremely important for solving the same problem many times with different values. In animation and film-making, programmers use complex equations to make action figures move like real people. Most modern appliances are now smart devices with responses that rely on the use of equations.



Shutterstock.com/NicoElNino

Chapter outline

	Working mathematically				
9.01 One-step equations	U	F		R	C
9.02 Two-step equations	U	F		R	C
9.03 Equations with variables on both sides*	U	F		R	C
9.04 Equations with brackets*	U	F		R	C
9.05 Simple quadratic equations $ax^2 = c$	U	F		R	C
9.06 Equation problems	U	F	PS	R	C
9.07 Equations and formulas	U	F	PS	R	C
9.08 Changing the subject of a formula*	U	F		R	C

*YEAR 9 EXTENSION

Wordbank

backtracking method A method of solving equations by 'undoing', or performing inverse (opposite) operations in reverse order

balancing method A method of solving equations by using the same operations on both sides of the equation

equation A mathematical statement that 2 quantities are equal, involving algebraic expressions and an equals sign (=)

inverse operation An opposite used in solving an equation, for example, the inverse operation of multiplying is dividing

quadratic equation An equation involving a variable squared, such as $3x^2 = 45$.

solve (an equation) To find the value of an unknown variable in an equation

solution The answer to an equation or problem, the correct value(s) of a variable that makes an equation true

substitute To replace a variable with a number

variable A quantity that can take on different values, represented by a symbol such as a letter of the alphabet

Quiz
Wordbank 9

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (10):

9.01 One-step equations
9.02 Two-step equations • Solving equations
9.03 Equations with variables on both sides • Solving equations
9.04 Equations with brackets • Solving equations
9.05 Simple quadratic equations $ax^2 = c$
9.06 Equation problems • Solving equations
9.07 Equations and formulas 1 • Equations and formulas 2
9.08 Changing the subject of a formula

Twig video (1):

9.01 The Arabic science of balancing

PhET interactives (2):

9.01 Equality explorer: Basics • Equality explorer

Quizzes (5):

- Wordbank 9
- SkillCheck 9
- Mental skills 9
- Language of maths 9
- Test yourself 9

Skillsheets (3):

9.02 Solving equations by balancing • Solving equations by backtracking • Solving equations using diagrams

Worksheets (16):

9.01 Equations 1 • One-step equations • Guess-and-check
9.02 Backtracking • Equations writing activity
9.03 Equations with unknowns on both sides
9.04 Equations 2 • Writing equations • Equations 3 (Extension) • Checking solutions
9.06 Equation problems • Solving equations review • Angle problems with algebra
9.07 Working with formulas
9.08 Changing the subject of a formula
Language of maths Writing equations
Mind map Equations and inequalities

Puzzles (10):

9.01 Equations dominoes • One-step equations
9.02 Solving equations
9.03 Equations with unknowns on both sides
9.04 Equations match • Equations • Equations order activity
9.06 Writing and solving equations
9.08 Formulas squaresaw
Language of maths Equations crossword

Presentations (3):

9.03 Formally solving equations
9.06 Writing equations • Applying linear equations

Technology (1):

9.02 Solving linear equations

Spreadsheet (1):

9.02 Linear equation solver

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- solve one- and two-step equations
- (EXTENSION, YEAR 9 CORE) solve equations involving variables on both sides, and brackets
- solve problems by using equations
- solve simple quadratic equations of the form $ax^2 = c$
- solve equations involving formulas to solve problems
- (EXTENSION, YEAR 9 PATH) change the subject of a formula

SkillCheck ANSWERS ON P. 599

1 Find the number represented by $\square$ to make each equation true.

a $\square + 3 = 10$ **b** $4 \times \square = 28$ **c** $12 - \square = 4$

d $\square \div 3 = 9$ **e** $9 + \square = 15$ **f** $\square - 7 = 16$

g $\square \times 6 = 30$ **h** $24 \div \square = 8$ **i** $10 + \square = 20$

j $\square - 5 = 0$ **k** $\frac{1}{2} \times \square = 4$ **l** $\square \div 5 = 4$

2 Evaluate each expression.

a $5 - 8$ **b** $-8 + 3$ **c** $3 \times (-4)$

d $-6 \div 3$ **e** $-2 + 5$ **f** $-2 - 9$

g $-4 \times (-9)$ **h** $-12 \div (-4)$ **i** $-3 - 7$

j $80 \div (-10)$ **k** $6 - 15$ **l** -8×9

3 If $d = 7$, evaluate:

a $2d + 3$ **b** $3d - 4$ **c** $\frac{d+1}{2}$ **d** $18 - 2d$

4 If $x = -3$, evaluate:

a $5x + 9$ **b** $4 - x$ **c** $\frac{x-3}{-1}$ **d** $2x - 8$

5 What is the opposite operation to:

a adding? **b** dividing? **c** multiplying? **d** subtracting?

Quiz
SkillCheck 9

One-step equations 9.01

An **equation** is a statement involving a variable (such as x), numbers and an equals (=) sign, for example, $\mathbf{2x - 5 = 11}$. In Year 7 we learnt how to **solve** equations, that is, to find the value of the variable that makes the equation true. This value is called the **solution** to the equation.

There are 2 algebraic methods for solving equations.

The **balancing method** involves representing both sides of an equation as balance scales, and solving the equation by performing the same operation on both sides to keep it 'balanced'.

The **backtracking method** involves using a flow chart to show what operations are performed on the variable (say, x) to create the equation, then using a reverse flow chart to undo each operation by performing the inverse (opposite) operation in reverse order.

Worksheets
Equations 1
One-step equations
Guess-and-check

Puzzles
Equations dominoes
One-step equations

Solving equations

To solve an equation, aim to have the variable (such as x) on one side of the equation and a number on the other side, in the form:

$$x = ____$$

Check your solution by substituting it back into the equation.

Videos
One-step equations

The Arabic science of balancing

Interactives
Equality explorer: Basics

Equality explorer

Example 1

Solve each equation.

a $m + 4 = 12$ **b** $3y = 18$ **c** $k - 3 = 5$ **d** $\frac{t}{8} = 6$

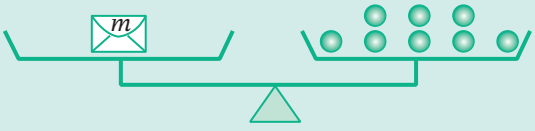
Solution

a **Method 1: The balancing method**

$m + 4 = 12$ This equation can be represented on a balance scale:



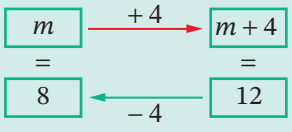
$m + 4 - 4 = 12 - 4$ Subtracting 4 from both sides.



$m = 8$ Check: $8 + 4 = 12$.

Method 2: The backtracking method

$m + 4 = 12$ This equation can be represented on a flowchart, starting at m and ending at 12:



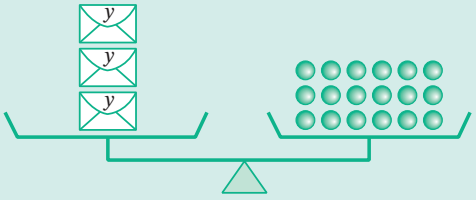
$m = 12 - 4$ Undo '+ 4' by subtracting 4.

Starting at 12 on the flowchart and ending at 8.

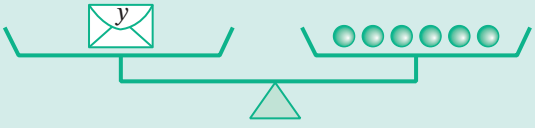
$m = 8$ Check: $8 + 4 = 12$.

b **Method 1: The balancing method**

$3y = 18$



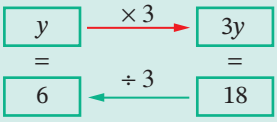
$\frac{3y}{3} = \frac{18}{3}$ Dividing both sides by 3.



$y = 6$ Check: $3 \times 6 = 18$

Method 2: The backtracking method

$3y = 18$



$y = \frac{18}{3}$ Undo '× 3' in $3y$ by dividing by 3.

$y = 6$ Check: $3 \times 6 = 18$

c **Method 1: The balancing method**

$k - 3 = 5$

$k - 3 + 3 = 5 + 3$ Adding 3 to both sides.

$k = 8$ Check: $8 - 3 = 5$

Method 2: The backtracking method

$k - 3 = 5$

$k = 5 + 3$ Undo '− 3' by adding 3.

$k = 8$ Check: $8 - 3 = 5$

d **Method 1: The balancing method**

$\frac{t}{8} = 6$

$\frac{t}{8} \times 8 = 6 \times 8$ Multiplying both sides by 8.

$t = 48$ Check: $\frac{48}{8} = 6$

Method 2: The backtracking method

$\frac{t}{8} = 6$

$t = 6 \times 8$ Undo '÷ 8' by multiplying by 8.

$t = 48$ Check: $\frac{48}{8} = 6$

These are all 'one-step' equations because they require only one step to solve.

Inverse operations

Operation	Inverse operation
+	−
−	+
×	÷
÷	×

EXERCISE 9.01 ANSWERS ON P. 599

One-step equations U F R C

EXAMPLE 1

1 Solve each equation, showing the working.

a	$x + 5 = 12$	**b**	$y + 13 = 36$	**c**	$m + 9 = 21$	**d**	$k + 27 = 54$
e	$k + 6 = 2$	**f**	$s + 3 = 1$	**g**	$n + 9 = -4$	**h**	$p + 17 = -3$
i	$m + 8 = 0$	**j**	$g + 4.5 = 20$	**k**	$a + 1\frac{3}{4} = 12$	**l**	$k + 2.7 = -5$

2 Solve each equation, showing the working.

a	$t - 11 = 40$	**b**	$w - 7 = 4$	**c**	$a - 5 = 2$	**d**	$j - 23 = 51$
e	$q - 12 = 17$	**f**	$f - 42 = 68$	**g**	$x - 9 = 24$	**h**	$g - 10 = 1$
i	$j - 3 = -5$	**j**	$w - 9 = -2$	**k**	$m - 12 = -27$	**l**	$d - 1 = -1$
m	$y - 17 = 3.9$	**n**	$n - 2.1 = 5.9$	**o**	$b - \frac{4}{5} = \frac{1}{4}$	**p**	$s - 2\frac{1}{3} = 4\frac{1}{2}$

3 Solve each equation, showing the working.

a	$6x = 24$	**b**	$4g = 16$	**c**	$10y = 90$	**d**	$5b = 45$
e	$12d = 108$	**f**	$9c = 81$	**g**	$23p = 115$	**h**	$2m = 26$
i	$5x = -25$	**j**	$7q = -42$	**k**	$-3c = 54$	**l**	$-15x = -120$
m	$4p = 37$	**n**	$8r = 18$	**o**	$3q = -10$	**p**	$9w = -12$

4 Solve each equation, showing the working.

a	$\frac{s}{3} = 5$	**b**	$\frac{m}{7} = 6$	**c**	$\frac{d}{10} = 12$	**d**	$\frac{w}{2} = 29$
e	$\frac{a}{3} = -4$	**f**	$\frac{f}{-2} = 9$	**g**	$\frac{r}{5} = -3$	**h**	$\frac{m}{-7} = 1$
i	$\frac{x}{-8} = -6$	**j**	$\frac{t}{2} = \frac{3}{4}$	**k**	$\frac{n}{-4} = -4$	**l**	$\frac{x}{3} = -\frac{1}{6}$

5 Solve $3x = -12$. Select the correct answer **A**, **B**, **C** or **D**.

A	$x = -9$	**B**	$x = -36$	**C**	$x = -4$	**D**	$x = -15$

6 Write a one-step equation whose solution is: R C

a	$m = 4$	**b**	$x = 0$	**c**	$r = 7$	**d**	$d = -5$

Foundation Standard Complex

9780170479516

Two-step equations

9.02

Two-step equations require 2 steps or 2 inverse operations to solve.

Example 2

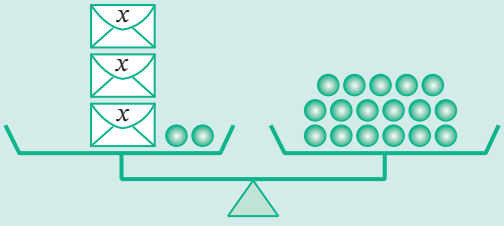
Solve each equation.

a $3x + 2 = 17$ **b** $\frac{k}{5} + 4 = 7$

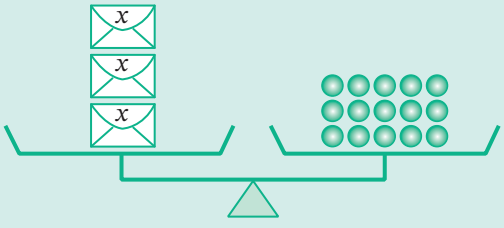
Solution

a **Method 1: The balancing method**

$3x + 2 = 17$

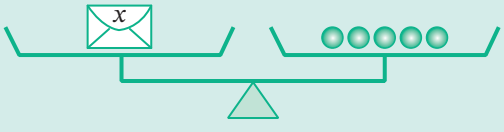


$3x + 2 - 2 = 17 - 2$ *Step 1*: Subtracting 2 from both sides.



$3x = 15$

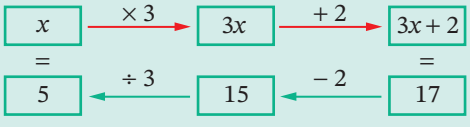
$\frac{3x}{3} = \frac{15}{3}$ *Step 2*: Dividing both sides by 3.



$x = 5$ Check: $3 \times 5 + 2 = 17$

Method 2: The backtracking method

$3x + 2 = 17$



Videos
Two-step equations
Solving equations

Worksheets
Backtracking
Equations writing activity

Skillsheets
Solving equations by balancing
Solving equations by backtracking
Solving equations using diagrams

Technology
Solving linear equations

Spreadsheet
Linear equation solver

Puzzle
Solving equations

$3x = 17 - 2$ *Step 1*: Undo '+ 2' by subtracting 2.

$x = \frac{15}{3}$ *Step 2:* Undo '× 3' in 3x by dividing by 3.

$x = 5$ Check: $3 \times 5 + 2 = 17$

b **Method 1: The balancing method**

$\frac{k}{5} + 4 = 7$

$\frac{k}{5} + 4 - 4 = 7 - 4$ *Step 1*: Subtracting 4 from both sides.

$\frac{k}{5} = 3$

$\frac{k}{5} \times 5 = 3 \times 5$ *Step 2*: Multiplying both sides by 5.

$k = 15$ Check: $\frac{15}{5} + 4 = 3 + 4 = 7$

Method 2: The backtracking method

$\frac{k}{5} + 4 = 7$

$\frac{k}{5} = 7 - 4$ *Step 1*: Undo '+ 4' by subtracting 4.

$\frac{k}{5} = 3$

$k = 3 \times 5$ *Step 2*: Undo '÷ 5' by multiplying by 5.

$k = 15$ Check: $\frac{15}{5} + 4 = 3 + 4 = 7$

Video
Two-step equations

Example 3

Solve each equation.

a $\frac{2h}{3} = 4$ **b** $\frac{x+1}{2} = 7$

Solution

a **Method 1: The balancing method**

$\frac{2h}{3} = 4$

$\frac{2h}{3} \times 3 = 4 \times 3$ *Step 1*: Multiplying both sides by 3.

$2h = 12$

$\frac{2h}{2} = \frac{12}{2}$ *Step 2*: Dividing both sides by 2.

$h = 6$ Check: $\frac{2 \times 6}{3} = 4$

Method 2: The backtracking method

$\frac{2h}{3} = 4$

$2h = 4 \times 3$ *Step 1*: Undo '÷ 3' by multiplying by 3.

$2h = 12$

$h = \frac{12}{2}$ *Step 2*: Undo '× 2' in 2h by dividing by 2.

$h = 6$ Check: $\frac{2 \times 6}{3} = 4$

b **Method 1: The balancing method**

$\frac{x+1}{2} = 7$

$\frac{x+1}{2} \times 2 = 7 \times 2$ *Step 1*: Multiplying both sides by 2.

$x + 1 = 14$

$x + 1 - 1 = 14 - 1$ *Step 2*: Subtracting 1 from both sides.

$x = 13$ Check: $\frac{13+1}{2} = 7$

Method 2: The backtracking method

$\frac{x+1}{2} = 7$

$x + 1 = 7 \times 2$ *Step 1*: Undo '÷ 2' by multiplying by 2.

$x + 1 = 14$

$x = 14 - 1$ *Step 2*: Undo '+ 1' by subtracting 1.

$x = 13$ Check: $\frac{13+1}{2} = 7$

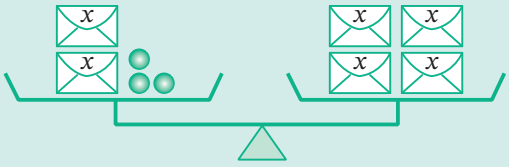
Example 4

Solve $2x + 3 = 4x$.

Solution

This equation has the variable on **both sides** of the equation, so we need to use the **balancing method** to 'collect like terms' and simplify the equation.

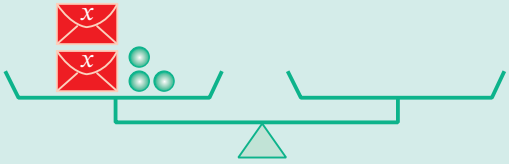
$2x + 3 = 4x$



$2x + 3 - 4x = 4x - 4x$

$-2x + 3 = 0$

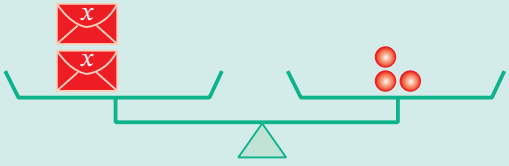
Subtracting $4x$ from both sides to remove it from the RHS (right-hand side).



$-2x + 3 - 3 = 0 - 3$

$-2x = -3$

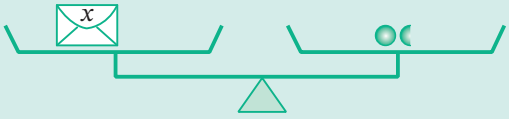
Subtracting 3 from both sides to remove it from the LHS (left-hand side).



$\frac{-2x}{-2} = \frac{-3}{-2}$

$x = 1\frac{1}{2}$

Divide both sides by (-2).



Check:

LHS $= 2\left(1\frac{1}{2}\right) + 3 = 6$

RHS $= 4\left(1\frac{1}{2}\right) = 6$ LHS = RHS

9780170479516

We can solve this equation in 2 steps if we move the variables to the RHS first by subtracting $2x$ from both sides.

$2x + 3 = 4x$

$3 = 2x$

$\frac{3}{2} = x$

$x = 1\frac{1}{2}$

EXERCISE 9.02 ANSWERS ON P. 599

Two-step equations U F R C

1 Check which equation has $p = -2$ as its solution. Select the correct answer **A**, **B**, **C** or **D**.

A $2 + 3p = 8$ **B** $5p - 2 = 8$ **C** $-4p + 16 = 8$ **D** $-3p + 2 = 8$

EXAMPLE 2

2 Solve each equation, showing all steps. Remember to check your answers.

a $2m + 7 = 19$ **b** $3x - 5 = 13$ **c** $5k + 12 = 52$

d $6w - 17 = 19$ **e** $4h + 21 = 9$ **f** $8d - 5 = -27$

g $-2x + 18 = 4$ **h** $-3a - 9 = -6$ **i** $14c + 2 = 37$

j $12p - 10 = 86$ **k** $9y + 11 = 101$ **l** $2r - 15 = 16$

3 Solve each equation.

a $\frac{m}{2} + 4 = 13$ **b** $\frac{c}{5} + 9 = 12$ **c** $\frac{d}{7} + 2 = 8$

d $\frac{k}{3} + 15 = 6$ **e** $\frac{h}{2} + 7 = 17$ **f** $\frac{x}{4} + 2 = -4$

g $\frac{n}{5} - 9 = 4$ **h** $\frac{t}{3} - 6 = -2$ **i** $\frac{a}{8} - 2 = -6$

j $\frac{x}{6} - 16 = -13$ **k** $\frac{v}{15} - 2 = 4$ **l** $\frac{b}{3} - 4 = 3$

4 This is Liam's incorrect solution for $7x + 5 = 13$.

$7x + 5 = 13$

$7x = 13 - 5$	**A**
$7x = 8$	**B**
$x = \frac{8}{7}$	**C**
$x = 1\frac{1}{8}$	**D**

In which line was the error made? Select **A**, **B**, **C** or **D**. R

EXAMPLE 3

5 Solve each equation.

a $\frac{3x}{2} = 9$ **b** $\frac{15x}{9} = 15$ **c** $\frac{2x}{3} = 8$ **d** $\frac{3N}{5} = 3$

e $\frac{6N}{7} = 18$ **f** $\frac{5B}{2} = -10$ **g** $\frac{3x}{4} = -3$ **h** $\frac{2x}{5} = -4$

i $\frac{5m}{2} = 11$ **j** $\frac{-x}{3} = 8$ **k** $\frac{-4x}{5} = 1$ **l** $\frac{-2x}{3} = -10$

6 Check which value of k is the solution to $\frac{k-12}{9} = 6$. Select **A**, **B**, **C** or **D**.

A $k = 27$ **B** $k = 42$ **C** $k = 66$ **D** $k = 162$

7 Solve each equation.

a $\frac{x+1}{3} = 2$ **b** $\frac{x-3}{2} = 3$ **c** $\frac{N+2}{5} = 1$ **d** $\frac{N-3}{4} = 5$

e $\frac{N+8}{5} = 6$ **f** $\frac{x+1}{2} = -2$ **g** $\frac{k-5}{2} = -5$ **h** $\frac{m+2}{3} = 11$

EXAMPLE 4

8 Solve each equation.

a $7x - 15 = 2x$ **b** $10p + 6 = 7p$ **c** $4r + 20 = r$ **d** $3u - 7 = 4u$

e $5d + 36 = 14d$ **f** $2y = y - 5$ **g** $9f = 4f + 11$ **h** $16 - 3a = a$

9 Write 4 equations, one of each type shown in questions **2**, **3**, **5**, **7** and **8**, that have the solution $p = 2$. R C

9.03 Extension: Equations with variables on both sides

YEAR 9 CORE



Worksheet Equations with unknowns on both sides

For equations with variables on both sides, such as $3x + 4 = 2x + 7$, we can only use the **balancing method**, not the backtracking method.

Equations with variables on both sides

Perform operations on both sides to move:

- all the variables onto one side of the equation
- all the numbers onto the other side of the equation.

Foundation Standard Complex

9780170479516

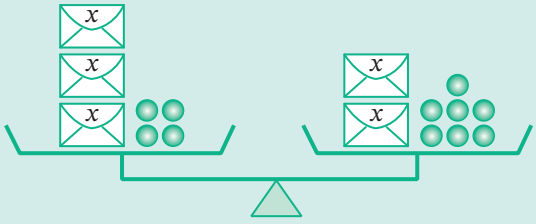
Example 5

Solve each equation.

a $3x + 4 = 2x + 7$ **b** $5n - 3 = 2n - 15$ **c** $2d - 8 = -5d - 71$

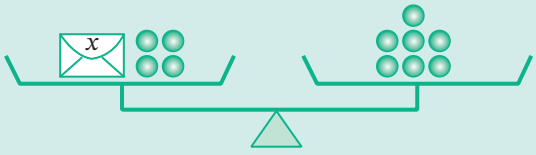
Solution

a $3x + 4 = 2x + 7$



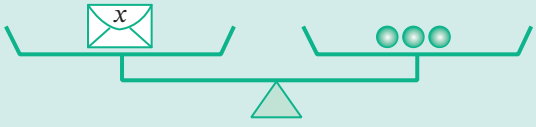
$3x + 4 - 2x = 2x + 7 - 2x$ Subtracting $2x$ from both sides to remove it from the RHS (right-hand side).

$x + 4 = 7$



$x + 4 - 4 = 7 - 4$ Subtracting 4 from both sides to remove it from the LHS (left-hand side).

$x = 3$



Check: LHS $= 3 \times 3 + 4 = 13$

RHS $= 2 \times 3 + 7 = 13$

LHS = RHS.

b $5n - 3 = 2n - 15$

$5n - 3 - 2n = 2n - 15 - 2n$ Subtracting $2n$ from both sides to remove it from the RHS.

$3n - 3 = -15$ Now this is a two-step equation.

$3n - 3 + 3 = -15 + 3$ Adding 3 to both sides to remove it from the LHS.

$3n = -12$ Now this is a one-step equation.

$\frac{3n}{3} = \frac{-12}{3}$ Dividing both sides by 3.

$n = -4$ Check: LHS $= 5 \times (-4) - 3 = -23$

RHS $= 2 \times (-4) - 15 = -23$ LHS = RHS

YEAR 9 CORE

Puzzle
Equations with unknowns on both sides

Presentation
Formally solving equations

Videos
Equations with variables on both sides

Solving equations

9.03

YEAR 9 CORE

c $2d - 8 = -5d - 71$

$2d - 8 + 5d = -5d - 71 + 5d$	Adding $5d$ to both sides to remove it from the RHS.
$7d - 8 = -71$	Now this is a two-step equation.
$7d - 8 + 8 = -71 + 8$	Add 8 to both sides to remove it from the LHS.
$7d = -63$	Now this is a one-step equation.
$\frac{7d}{7} = \frac{-63}{7}$	Dividing both sides by 7.
$d = -9$	Check: LHS $= 2 \times (-9) - 8 = -26$ RHS $= -5 \times (-9) - 71 = -26$ LHS = RHS.

EXERCISE 9.03 ANSWERS ON P. 600

Equations with variables on both sides U F R C

EXAMPLE 5

1 What is the first step in solving $3x - 4 = x + 5$? Select the correct answer **A**, **B**, **C** or **D**. R C

A subtract x from both sides **B** add x to both sides

C add $3x$ to both sides **D** subtract 4 from both sides

2 Solve $10m - 2 = 6m$. Select **A**, **B**, **C** or **D**.

A $m = -\frac{1}{2}$ **B** $m = \frac{1}{8}$ **C** $m = \frac{1}{2}$ **D** $m = 2$

3 Solve each equation, showing all steps. Remember to check your answers.

a $3a + 6 = a + 18$ **b** $2k + 4 = k + 8$ **c** $3x + 6 = 2x + 9$

d $6p + 4 = p + 19$ **e** $6n + 5 = 2n + 17$ **f** $5q + 6 = 4q + 12$

g $5y + 14 = 3y + 14$ **h** $4a + 7 = 2a + 21$ **i** $2r + 9 = r + 15$

4 Solve $4x - 5 = 2x + 7$. Select **A**, **B**, **C** or **D**.

A $x = 1$ **B** $x = 2$ **C** $x = 4$ **D** $x = 6$

5 Solve each equation.

a $5a - 3 = 2a + 6$ **b** $6x - 2 = 3x + 6$ **c** $3d - 6 = d - 4$

d $8p - 15 = 3p - 10$ **e** $3m + 5 = -m + 26$ **f** $4x + 6 = -6x + 56$

g $4x + 3 = -2x + 7$ **h** $2r + 18 = -5r + 11$ **i** $3y - 12 = -y + 6$

6 Here is Bree's solution for $3x - 4 = -2x + 8$.

$3x - 4 = -2x + 8$

$3x + 2x - 4 = 8$	Line 1
$5x - 4 = 8$	Line 2
$5x = 8 - 4$	Line 3
$5x = 4$	Line 4
$x = \frac{5}{4}$	Line 5

Foundation Standard Complex

9780170479516

In which lines did Bree make mistakes? Select **A**, **B**, **C** or **D**. R

A Lines 1 and 3 **B** Lines 2 and 5 **C** Lines 1 and 4 **D** Lines 3 and 5

7 Solve each equation.

a $3d + 4 = d$ **b** $10k = 12 - 8k$ **c** $5 - p = p + 9$

d $7 + x = 6x + 22$ **e** $6k - 11 = k$ **f** $3m + 20 = 7m - 2$

g $4t = 12 - 4t$ **h** $8j - 17 = 10j$ **i** $6 - 3q = 8 - q$

8 Write an equation with x on both sides that has the solution $x = 7$. R C

YEAR 9 CORE

9.04

Extension: Equations with brackets 9.04

Equations with brackets (grouping symbols)

Expand the expressions and then solve as usual.

Example 6

Solve each equation.

a $3(x + 5) = 9$ **b** $2(3 - y) = 4y + 4$ **c** $5(r - 3) = 2(r + 6)$

Solution

a $3(x + 5) = 9$

$3x + 15 = 9$ Expanding the expression to make it a two-step equation.

Can you think of another way to solve this equation without expanding?

$3x + 15 - 15 = 9 - 15$ Subtracting 15 from both sides.

$3x = -6$

$\frac{3x}{3} = \frac{-6}{3}$ Dividing both sides by 3.

$x = -2$ Check: $3(-2 + 5) = 3 \times 3 = 9$

b $2(3 - y) = 4y + 4$

$6 - 2y = 4y + 4$ Expanding the brackets.

$6 - 2y - 4y = 4y + 4 - 4y$ Subtracting $4y$ from both sides to remove it from the RHS.

$6 - 6y = 4$ Now this is a two-step equation.

$6 - 6y - 6 = 4 - 6$ Subtracting 6 from both sides to remove it from the LHS.

$-6y = -2$

YEAR 9 CORE

Worksheets
Equations 2
Writing equations
Equations 3 (Extension)
Checking solutions

Puzzles
Equations match
Equations
Equations order activity

Videos
Equations with brackets
Solving equations

Foundation Standard Complex

YEAR 9 CORE

$\frac{-6y}{-6} = \frac{-2}{-6}$ Dividing both sides by (-6).

$y = \frac{1}{3}$ Check: LHS $= 2 \times \left(3 - \frac{1}{3}\right) = 5\frac{1}{3}$

RHS $= 4 \times \frac{1}{3} + 4 = 5\frac{1}{3}$ LHS = RHS

c $5(r - 3) = 2(r + 6)$

$5r - 15 = 2r + 12$ Expanding both sides.

$5r - 15 - 2r = 2r + 12 - 2r$ Subtract $2r$ from both sides to remove it from the RHS.

$3r - 15 = 12$ Now this is a two-step equation.

$3r - 15 + 15 = 12 + 15$ Add 15 to both sides to remove it from the LHS.

$3r = 27$

$\frac{3r}{3} = \frac{27}{3}$ Dividing both sides by 3.

$r = 9$ Check: LHS $= 5 \times (9 - 3) = 5 \times 6 = 30$

RHS $= 2 \times (9 + 6) = 2 \times 15 = 30$ LHS = RHS

EXERCISE 9.04 ANSWERS ON P. 600

Equations with brackets UFRC

EXAMPLE 6

1 Solve each equation, showing all steps. Remember to check your answers.

a	$2(x + 3) = 8$	**b**	$3(m + 2) = 18$	**c**	$5(k + 1) = 35$
d	$4(p + 4) = 32$	**e**	$10(j + 2) = 50$	**f**	$7(d + 4) = 63$
g	$3(x - 4) = 15$	**h**	$3(x - 4) = 12$	**i**	$5(x - 1) = 10$
j	$8(x - 5) = 24$	**k**	$4(k - 5) = 28$	**l**	$2(q - 7) = 26$

2 Solve each equation.

a	$2(6 - x) = 6$	**b**	$3(4 - p) = 24$	**c**	$2(1 - q) = 8$
d	$3(16 - h) = 15$	**e**	$2(4 + r) = -8$	**f**	$3(e - 16) = -15$
g	$5(z - 3) = -10$	**h**	$-7(y + 2) = 35$	**i**	$9(6 - a) = 0$
j	$5(d + 3) = -20$	**k**	$-2(3 - k) = 12$	**l**	$4(6 - c) = -16$

3 Solve $2(12 - 3x) = 30$. Select the correct answer **A**, **B**, **C** or **D**.

A $x = -2$ **B** $x = -1$ **C** $x = 1$ **D** $x = 2$

4 Solve each equation.

a	$6(x + 1) = 3x + 3$	**b**	$2(r + 2) = r + 5$	**c**	$3(p + 3) = 2p + 2$
d	$2(x + 1) = x$	**e**	$5(p - 2) = 3p$	**f**	$4(e - 2) = 2e + 4$
g	$5(y + 2) = 3y + 12$	**h**	$3(a - 5) = 5 - 2a$	**i**	$3(2w + 1) = 3w - 15$

Foundation Standard Complex

9780170479516

5 Here is Ryan's solution for $7(h-4)=3h+20$.

$7(h-4)=3h+20$

$7h-4=3h+20$ **A**

$7h-4-3h=3h+20-3h$

$4h-4=20$ **B**

$4h-4+4=20+4$

$4h=24$ **C**

$\frac{4h}{4}=\frac{24}{4}$

$h=6$ **D**

In which line did Ryan make a mistake? Select **A**, **B**, **C** or **D**. R C

6 Solve each equation.

a $5(x-1)=4(x+2)$ **b** $6(x+2)=4(x+6)$ **c** $4(k+3)=3(k-2)$

d $7(y+2)=4(y+5)$ **e** $3(v+2)=2(v+5)$ **f** $4(x-2)=2(x+7)$

g $3(p+1)=5(p-1)$ **h** $3(2s+1)=5(s+2)$ **i** $5(d+4)=2(2d-1)$

7 Write an equation involving brackets whose solution is: R C

a $c=5$ **b** $k=-1$ **c** $n=2$ **d** $q=-4$

Mental skills 9: Maths without calculators ANSWERS ON P. 600

Quiz
Mental skills 9

Multiplying and dividing by 5, 15, 25 and 50

It is easier to multiply or divide a number by 10 than by 5. So whenever we multiply or divide a number by 5, we can double the 5 (to make 10) and then adjust the first number.

1 Study each example.

a To multiply by 5, halve the number, then multiply by 10.

$18\times5=18\times\frac{1}{2}\times10$ (or $9\times2\times10$)

$=9\times10$

$=90$

b To multiply by 50, halve the number, then multiply by 100.

$26\times50=26\times\frac{1}{2}\times100$ (or $13\times2\times100$)

$=13\times100$

$=1300$

c To multiply by 25, quarter the number, then multiply by 100.

$44\times25=44\times\frac{1}{4}\times100$ (or $11\times4\times25$)

$=11\times100$

$=1100$

d To multiply by 15, halve the number, then multiply by 30.

$8\times15=8\times\frac{1}{2}\times30$ (or $4\times2\times15$)

$=4\times30$

$=120$

Foundation Standard Complex

e To divide by 5, divide by 10 and double the answer. We do this because there are two 5s in every 10.

$140 \div 5 = 140 \div 10 \times 2$
$= 14 \times 2$
$= 28$

f To divide by 50, divide by 100 and double the answer. This is because there are two 50s in every 100.

$400 \div 50 = 400 \div 100 \times 2$
$= 4 \times 2$
$= 8$

g To divide by 25, divide by 100 and multiply the answer by 4. This is because there are four 25s in every 100.

$600 \div 25 = 600 \div 100 \times 4$
$= 6 \times 4$
$= 24$

h To divide by 15, divide by 30 and double the answer. This is because there are two 15s in every 30.

$240 \div 15 = 240 \div 30 \times 2$
$= 8 \times 2$
$= 16$

2 Now evaluate each expression.

a	32×5	**b**	14×5	**c**	48×5	**d**	18×50
e	52×50	**f**	36×25	**g**	28×5	**h**	12×25
i	12×15	**j**	22×35	**k**	$90 \div 5$	**l**	$170 \div 5$
m	$230 \div 5$	**n**	$1300 \div 50$	**o**	$900 \div 50$	**p**	$300 \div 25$
q	$1000 \div 25$	**r**	$360 \div 45$	**s**	$210 \div 15$	**t**	$360 \div 15$

Investigation

Solving $x^2 = c$

1 What is the inverse operation of 'squaring'?

2 The equation $x^2 = 9$ has 2 solutions. What are the 2 numbers, x, which when squared give 9?

3 What are the solutions for each equation? What is the pattern found in the answers?

a $x^2 = 25$ **b** $x^2 = 100$ **c** $x^2 = 1$

4 Study this example:

$x^2 = 49$
$x = \pm\sqrt{49}$ which means $x = \sqrt{49}$ or $-\sqrt{49}$
$= \pm 7$ which means $x = 7$ or $x = -7$.

Check: When $x = 7$, $x^2 = 7^2 = 49$

When $x = -7$, $x^2 = (-7)^2 = 49$

Use the same method to solve each equation and check your answers:

a $x^2 = 81$ **b** $x^2 = 64$

Simple quadratic equations $ax^2 = c$ 9.05

An equation involving a variable squared, such as $x^2 = 25$ or $3x^2 - 4 = 7$, is called a **quadratic equation**. In this section, we will solve simple quadratic equations of the type $ax^2 = c$, where c is a number.



Video
Simple quadratic equations $ax^2 = c$

Example 7

Solve each quadratic equation.

a $x^2 = 36$

b $x^2 = 121$

c $x^2 = 40$, writing the solution correct to one decimal place

d $x^2 = 83$, writing the solution as a surd

Solution

a
$x^2 = 36$
$x = \pm\sqrt{36}$ Finding the square root of both sides.
$= \pm 6$ The equation has 2 solutions, $x = 6$ or $x = -6$.

b
$x^2 = 121$
$x = \pm\sqrt{121}$ Finding the square root of both sides.
$= \pm 11$ The equation has solutions $x = 11$ or $x = -11$.

c
$x^2 = 40$
$x = \pm\sqrt{40}$
$= \pm 6.3345\ldots$
$\approx \pm 6.3$ The equation has solutions $x \approx 6.3$ or $x \approx -6.3$.

d
$x^2 = 83$ Leaving the answer as a surd.
$x = \pm\sqrt{83}$ $x = \sqrt{83}$ or $-\sqrt{83}$

Simple quadratic equations $x^2 = c$

The simple quadratic equation $x^2 = c$ (where c is a positive number) has 2 solutions,

$$x = \pm\sqrt{c}$$

(which means $x = \sqrt{c}$ or $x = -\sqrt{c}$).

Example 8

Solve $5x^2 = 125$.

Solution

$5x^2 = 125$

$x^2 = 25$ Divide both sides by 5 first.

$x = \pm\sqrt{25}$

$= \pm 5$ The equation has solutions $x = 5$ or $x = -5$.

EXERCISE 9.05 ANSWERS ON P. 600

Simple quadratic equations $ax^2 = c$ U F R C

EXAMPLE 7

1 Solve each quadratic equation.

a $x^2 = 81$ **b** $x^2 = 144$ **c** $x^2 = 1$

d $x^2 = 169$ **e** $m^2 = 5041$ **f** $u^2 = 1849$

2 Solve each quadratic equation, writing the solution correct to 2 decimal places.

a $x^2 = 13$ **b** $x^2 = 54$ **c** $x^2 = 88$

d $t^2 = 129$ **e** $h^2 = 946$ **f** $z^2 = 527$

3 Solve each quadratic equation, writing the solution as a surd.

a $x^2 = 41$ **b** $x^2 = 30$ **c** $x^2 = 48$

d $a^2 = 126$ **e** $p^2 = 75$ **f** $b^2 = 509$

4 Many simple quadratic equations have 2 solutions (one positive, one negative), but there is one simple quadratic equation that has only **one** solution. What is this equation and what is its solution? R

5 The equation $x^2 = -100$ has **no solutions**. Explain why. R C

6 Write 2 more simple quadratic equations that have no solutions. R C

7 Copy and complete: R C

a $x^2 = c$ has 2 solutions if c is _____.

b $x^2 = c$ has no solutions if c is _____.

c $x^2 = c$ has one solution if c is _____.

EXAMPLE 8

8 Solve each quadratic equation. R

a $3x^2 = 27$ **b** $2x^2 = 32$ **c** $3x^2 = 75$ **d** $5x^2 - 1 = 19$

e $5x^2 + 1 = 21$ **f** $7 + 3x^2 = 34$ **g** $6x^2 - 8 = 142$ **h** $7x^2 + 9 = 261$

Did you know?

Applications of quadratic equations

Quadratic equations can be more complex, for example, $3x^2 + 4x - 15 = 0$, with applications in many areas:

- the shape of satellite dishes and reflecting telescopes
- the path of a thrown object, such as a ball, shotput or javelin
- the orbits of planets as they move around the Sun.

Research the formula that solves quadratic equations.



Shutterstock.com/Dabarti CGI

Equation problems

9.06

Solving word problems requiring an equation

- Choose your variable
- Translate the words into an equation
- Solve the equation
- Write a sentence that answers the problem

Example 9

a When a number is doubled and 11 is subtracted, the result is 23. Find the number.

b 5 times a number is the same as 6 more than 3 times the number. What is the number?

Solution

a Let the number be x.

$x \times 2 - 11 = 23$ Translating the words into an equation

$2x - 11 = 23$

Solving the equation:

$2x - 11 + 11 = 23 + 11$ Adding 11 to both sides

$2x = 34$

$\frac{2x}{2} = \frac{34}{2}$ Dividing both sides by 2

$x = 17$ Check: $2 \times 17 - 11 = 23$

The number is 17.

Worksheets
Equation problems
Solving equations review
Angle problems with algebra

Puzzle
Writing and solving equations

Presentations
Writing equations
Applying linear equations

Videos
Equation problems
Solving equations

b Let n represent the number.

$$5n = 3n + 6$$

$5n - 3n = 3n + 6 - 3n$ Subtracting $3n$ from both sides

$$2n = 6$$

$\frac{2n}{2} = \frac{6}{2}$ Dividing both sides by 2

$$n = 3$$

The number is 3. Check: LHS $= 5 \times 3 = 15$

RHS $= 3 \times 3 + 6 = 15$ LHS = RHS

Example 10

Cooper charges for fixing washing machines using the formula: $C = 45h + 70$, where C is the charge in dollars and h is the number of hours the job takes. Find:

a the charge for a job that takes 3 hours

b the number of hours Cooper worked if the charge is \$430.

Solution

a Substitute $h = 3$ into the formula.

$$C = 45h + 70$$
$$= 45 \times 3 + 70$$
$$= 205$$

The charge is \$205.

b Substitute $C = 430$ into the formula and solve the equation.

$$430 = 45h + 70$$

$$45h + 70 = 430$$

$45h + 70 - 70 = 430 - 70$

$$45h = 360$$

$\frac{45h}{45} = \frac{360}{45}$

$$h = 8$$

The number of hours Cooper worked is 8.



Shutterstock.com/Africa Studio

EXERCISE 9.06 ANSWERS ON P. 600

Equation problems U F PS R C

EXAMPLE 9

1 Solve each problem by writing an equation and then solving it. You may use diagrams to help you think about the information. R C

a 4 family tickets for a theme park cost $352. How much does each ticket cost? (Let t represent the price of a family ticket.)

b 20 cans of drink cost $36. How much does each can cost? (Let c represent the cost of one can.)

c A number is tripled and the result is 99. What is the number? (Let x represent the unknown number.)

d A number has 7 added to it and the result is 23. Find the number. (Let n represent the unknown number.)

2 Select the correct equation **A**, **B**, **C** or **D** for each problem. Then use the equation to solve the problem. R C

a If 23 is subtracted from a number, the answer is 79. What is the number?

A $79 + N = 23$ **B** $23N = 79$ **C** $23 - N = 79$ **D** $N - 23 = 79$

b The sum of a number and 18 is 67. What is the number?

A $N + 67 = 18$ **B** $N + 18 = 67$ **C** $18N = 67$ **D** $N - 18 = 67$

c When a number is multiplied by 23, the answer is 1288. What is the number?

A $1288N = 23$ **B** $N + 23 = 1288$ **C** $1288 - N = 23$ **D** $23N = 1288$

Foundation Standard Complex

d When a number is divided by 14, the answer is 42. What is the number?

A $N - 14 = 42$ **B** $\frac{N}{14} = 42$ **C** $14N = 42$ **D** $14 + N = 42$

e Marika buys a car that costs $13 499. She pays a deposit of $4525. How much does she owe?

A $4525N = 13\,499$ **B** $N - 4525 = 13\,499$

C $N + 4525 = 13\,499$ **D** $N = 13\,499 + 4525$

3 For each problem, write an equation and find the unknown number. R C

a When an unknown number is divided by 5, the result is 15.

b When 12 is subtracted from a number, the answer is 37.

c When a number is subtracted from 52, the answer is 25.

d An unknown number is multiplied by 3, then 8 is subtracted. The result is 40.

e An unknown number is halved, then 7 is added. The result is 13.

f The sum of a number and 4 is doubled. The result is 44.

g The sum of a number and 6 is divided by 3 and the result is 7.

h The product of a number and 5 is decreased by 2. The result is 28.

4 Translate each problem into an equation, then solve the equation to solve the problem. PS R C

a The student council is holding a disco to raise money. Each ticket bought by students raises $20, but the costs of running the disco total $1000. How many tickets must be sold to make a profit of $2000? (Let n stand for the number of tickets sold.)

b In 8 years Kelly's age will be twice what it is now. How old is Kelly now? (Let n stand for Kelly's age now.)

c 8 sheep have the same mass as 3 sheep and one cow. If the cow's mass is 500 kg, what is the mass of one sheep? (Let s stand for the mass of one sheep.)

d If you multiply Liam's favourite number by 3 and add 1, you get the same answer as if you multiplied the number by 5 and took away 11. What is the number? (Let x represent the number.)

e Mr Yen says, 'If you add 15 to my age and multiply by 7, the answer is 483.' How old is Mr Yen? (Let a stand for his age.)

f The area of a rhombus is calculated by multiplying its diagonals and dividing by 2. If a rhombus has an area of 88 cm^2 and one diagonal is 11 cm, what is the length of the other diagonal? (Let d represent the length of the diagonal.)

EXAMPLE 10

5 A temperature in degrees Celsius (°C) can be converted to degrees Fahrenheit (°F) using the formula $F = \frac{9C}{5} + 32$. PS R

a Convert 25°C to °F.

b Convert 108°F to °C.

6 The charge, $C, for hiring a hall for an event is $C = 150 + 2N$, where N stands for the number of people at the event. Find: PS R

a the charge when 225 people are at the event

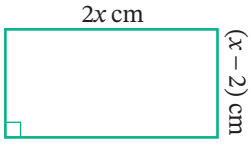
b the number of people at the event when the charge is $394.

7 The cost, $y, of an online ad on a local news website is $y = 0.8w + 3.5$, where w is the number of words in the ad. Find: PS R

a the cost of a 13-word ad

b the number of words in an ad costing $25.90

8 The perimeter of this rectangle is 47 cm. Find: PS R



a the value of x

b the length of the rectangle

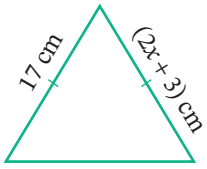
c the width of the rectangle.

9 The profit, $P, made by a hairdresser is given by $P = 18x - 900$, where x represents the number of customers. Find: PS R

a the profit made with 195 customers

b the number of customers if the profit is $3060.

10 Find the value of x in this isosceles triangle. R



9.07

Equations and formulas 9.07

A **formula** is a rule written in algebraic form. It shows the relationship between variables.

For example:

- the formula $A = lw$ gives the area, A, of a rectangle with length l and width w.
- the formula $D = ST$ gives the distance, D, travelled by a car at a speed S for time T.

Solving mathematical problems often involves substituting values in formulas and solving equations.

Videos
Equations and formulas 1
Equations and formulas 2

Worksheet
Working with formulas

Example 11

The formula for the perimeter, P, of a rectangle of length l and width w is given by $P = 2(l + w)$. Find a rectangle's:

a length if its perimeter is 40 cm and its width is 8 cm

b width if its perimeter is 100 cm and its length is 35 cm.

Foundation Standard Complex

Solution

a Substitute $P = 40$, $w = 8$:

$$P = 2(l + w)$$
$$40 = 2(l + 8)$$
$$40 = 2l + 16$$
$$2l + 16 = 40$$
$$2l + 16 - 16 = 40 - 16$$
$$2l = 24$$
$$\frac{2l}{2} = \frac{24}{2}$$
$$l = 12$$

b Substitute $P = 100$, $l = 35$:

$$P = 2(l + w)$$
$$100 = 2(35 + w)$$
$$100 = 70 + 2w$$
$$70 + 2w = 100$$
$$70 + 2w - 70 = 100 - 70$$
$$2w = 30$$
$$\frac{2w}{2} = \frac{30}{2}$$
$$w = 15$$

EXERCISE 9.07 ANSWERS ON P. 600

Equations and formulas U F PS R C

EXAMPLE 11

1 The volume of a rectangular prism with length, l, width, w, and height, h, is given by $V = lwh$. Use this formula to find the length of a rectangular prism with width 4 and height 5 if its volume is 340. Select the correct answer **A**, **B**, **C** or **D**.

A 17 **B** 68 **C** 85 **D** 425

2 The number of toothpicks (T) needed to build a row of N squares is $T = 3N + 1$.

a How many toothpicks are needed to build a row of 4 squares?

b How many squares can be built using 322 toothpicks?

3 The number of hours (H) of sleep that children need depends on their age (A) in years and is given by the formula $H = 17 - \frac{A}{2}$. Find: PS R

a the hours of sleep needed when a child is 14 years old

b how old a child who needs 8 hours sleep is.

4 Using the perimeter formula for a rectangle, $P = 2(l + w)$, find: PS R

a the perimeter when length = 13 and width = 4

b the length when perimeter = 36 and width = 7

c the width when perimeter = 58 and length = 16

5 The area of a trapezium is given by $A = \frac{1}{2}(a+b)h$, where a and b are the lengths of its parallel sides and h is the perpendicular height between them. Find the value of: PS R

a the area of a trapezium with parallel sides 20 and 13 and height 8

b the height of a trapezium with parallel sides 22 and 8 and area 90

c a parallel side of a trapezium with area 64, height 4 and other parallel side 17

d a parallel side of a trapezium with other parallel side 5, height 12 and area 120

Foundation Standard Complex

9780170479516

6 The volume of a pyramid is given by $V = \frac{1}{3}Ah$, where A is the area of the base and h is the height of the pyramid. Find the value of: PS R

- **a** the volume of a pyramid with base area 25 and height 7
- **b** the base area of a pyramid with volume 100 and height 10
- **c** the height of a pyramid with volume 225 and base area 75

7 The simple interest (I) earned on principal \$$P$ invested for n years at an interest rate of r p.a. (where r is a decimal) is given by $I = Prn$. PS R

p.a. means 'per year'

- **a** Find the simple interest earned when \$1250 is invested at 2% p.a. for 5 years (convert 2% to a decimal first).
- **b** Find the principal invested if interest was \$60 and the interest rate was 5% p.a. for 3 years.
- **c** Find the interest rate (as a percentage) if \$700 invested for 3 years earned \$84 in interest.
- **d** Find the number of years that \$1650 was invested if it earned \$346.50 interest at 3% p.a.

8 **a** 3 people meet and each person shakes hands with every other person. How many handshakes were there? R

- **b** In how many ways can 4 people shake hands?
- **c** The formula $H = \frac{P(P-1)}{2}$, where H is the number of handshakes, and P is the number of people, gives the number of shakes between people. In how many ways can 10 people shake hands?
- **d** How many people are needed for 105 handshakes?

Extension: Changing the subject of a formula

9.08

The formula $C = 2\pi r$ is used to calculate the circumference of a circle if we know the value of r, the length of the circle's radius. The formula is used to find C, and so C is called the **subject** of the formula. We could rewrite the formula so that r was the subject and on the LHS while C is on the RHS. This is called **changing the subject of the formula**. To change the subject of a formula, use the same rules as for solving an equation. The answer is not a number but an algebraic expression (another formula).

YEAR 9 PATH

Worksheet
Changing the subject of a formula

Puzzle
Formulas squaresaw

Foundation Standard Complex

YEAR 9 PATH

Example 12

Rearrange the formula $C = 2\pi r$ to make r the subject.

Solution

To make r the subject of the formula, solve it like an equation for r.

$C = 2\pi r$

$2\pi r = C$ Swapping sides so that r appears on the LHS.

$r = \frac{C}{2\pi}$ Dividing both sides by 2π.

Example 13

Video
Changing the subject of a formula

Change the subject of the formula $v = u + at$ to a.

Solution

$v = u + at$

$u + at = v$ Swapping sides so that a appears on the LHS.

$at = v - u$ Subtracting u from both sides.

$a = \frac{v-u}{t}$ Divide both sides by t.

EXERCISE 9.08 ANSWERS ON P. 600

Changing the subject of a formula U F R C

EXAMPLE 12

1 Rearrange each formula to make the variable in brackets the subject. R C

If a is the variable in brackets, the answer should be in the form '$a = \ldots$'.

a $F = ma\ (a)$ **b** $y = x + 3\ (x)$ **c** $A = \frac{d}{2}\ (d)$ **d** $f = \frac{120}{v}\ (v)$

2 The formula $y = 2x + 1$ is rearranged to make x the subject. What is the rearranged formula? Select the correct answer **A**, **B**, **C** or **D**. R C

A $x = y - 1 \div 2$ **B** $x = \frac{y}{2} - 1$ **C** $x = 2y - 1$ **D** $x = \frac{y-1}{2}$

EXAMPLE 13

3 Change the subject of the formula $p = ax - 2$ to x. R C

4 If $p = \frac{5q}{2}$, make q the subject. R C

5 The formula for converting temperatures from Celsius to Fahrenheit is $F = \frac{9C}{5} + 32$. Rearrange the formula to make C the subject. R C

6 The perimeter of a rectangle is $P = 2l + 2w$. Change the subject of the formula to w. R C

7 Change the subject of the formula $a = \frac{v-u}{t}$ to v. R C

Foundation Standard Complex

9780170479516

YEAR 9 PATH

9.08

8 The average speed S km/h of a car travelling D kilometres in T hours is given by the formula $S = \frac{D}{T}$. R C

a Change the subject to T.

b Find the time taken for a car to travel 600 km at 75 km/h.

9 Pythagoras' theorem is $c^2 = a^2 + b^2$. Change the subject of the formula to a. R C

10 Make d the subject of the formula $x = 2(d - 5)$. R C

11 If the volume of a cylinder is $V = \pi r^2 h$, change the subject to the radius r. R C

12 The area of a trapezium is $A = \frac{1}{2}(a + b)h$. Change the subject of the formula to b. R C

Power plus ANSWERS ON P. 601

1 Solve each equation.

a $2(x + 1) + 2(x - 1) = 12$

b $2(x + 4) - 3(x - 1) = 9$

c $4(2x - 1) - 5(x - 2) = 6$

d $2 - (3x + 5) = 4(x + 1)$

e $\frac{x+7}{4} = \frac{6(x-1)}{3}$

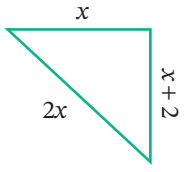
f $\frac{2(x+1)}{3} = \frac{5(x-2)}{2}$

g $\frac{2x}{3} - \frac{x}{6} = 10$

h $\frac{3x}{4} + \frac{9x}{10} = 44$

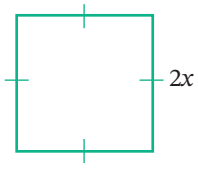
2 For each diagram, write an equation for x and solve it.

a



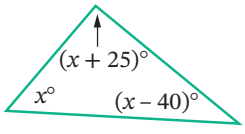
Perimeter = 30

b



Area = 100

c



3 Solve each quadratic equation.

a $4x^2 = 100$

b $5x^2 - 7 = 73$

c $\frac{2x^2}{3} = 24$

d $3x^2 + 8 = 20$

e $(x + 4)^2 = 9$

f $x^2 + 2x = 15$

4 Diophantus was a famous mathematician who was the first to abbreviate his mathematical thoughts using symbols. He is known as the Greek father of algebra and when he died, one of his admirers wrote the following riddle about his life:

Diophantus' youth lasted $\frac{1}{6}$ of his life. He grew a beard after $\frac{1}{12}$ more. After $\frac{1}{7}$ more of his life, Diophantus married; 5 years later he had a son. The son lived exactly $\frac{1}{2}$ as long as his father, and Diophantus died just 4 years after his son. All this adds to the years Diophantus lived.

Write this riddle as an equation and solve it to find how long Diophantus lived. (*Hint*: Let x years equal his life.)

Foundation Standard Complex

CHAPTER 9 REVIEW

Language of maths

Quiz
Language of maths 9

Puzzle
Equations crossword

Worksheet
Writing equations

backtracking	balancing	brackets	check
equation	expand	formula	inverse operation
LHS (left-hand side)	one-step equation	pronumeral	quadratic equation
RHS (right-hand side)	solution	solve	substitution
two-step equation	undoing	unknown	variable

1 What are the 2 algebraic methods for solving equations?

2 Which method involves 'undoing' operations?

3 What does the word 'solution' mean?

4 Which word in the list means 'opposite'?

5 Why is the **variable** in an equation sometimes called an **unknown**?

6 What word means to rewrite an algebraic expression by removing the brackets?

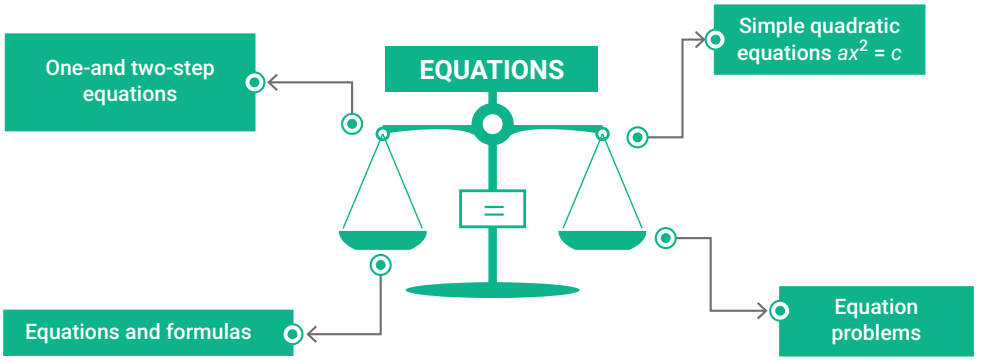
7 What does **RHS** stand for?

Topic summary

Worksheet
Mind map: Equations

- Which parts of this topic did you find easy? What did you already know?
- Give examples of some problems that might be solved using equations.
- Are there any parts of this topic that you still don't understand? Talk to your teacher about them.
- In what sort of careers would people use equations?

Copy and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



9780170479516

TEST YOURSELF 9

ANSWERS ON P. 601

Quiz
Test yourself 9

1 Solve each equation. 9.01

a $k + 6 = 13$ **b** $x - 3 = 8$ **c** $a + 3 = 17$

d $a - 12 = 21$ **e** $3x = 12$ **f** $10f = 120$

g $\frac{m}{4} = 7$ **h** $\frac{x}{3} = 8$ **i** $w + 9 = 3$

j $k - 5 = -7$ **k** $-2m = -16$ **l** $\frac{x}{3} = -12$

2 Solve each equation. 9.02

a $4p + 3 = 23$ **b** $3m + 17 = 8$ **c** $2x - 12 = 18$

d $\frac{h}{7} + 5 = 16$ **e** $\frac{n}{2} - 8 = -12$ **f** $\frac{a}{5} - 8 = 4$

g $\frac{2y}{5} = 4$ **h** $\frac{x+2}{3} = 4$ **i** $\frac{n-5}{2} = 10$

j $\frac{5x}{2} = -15$ **k** $7k = 2k + 130$ **l** $10d - 58 = 12d$

YEAR 9 PATH

3 Solve each equation. 9.03

a $3x + 4 = x + 6$ **b** $5u - 3 = 2u + 6$ **c** $12h - 8 = 8h + 4$

d $3v - 4 = 7v + 8$ **e** $2x + 9 = 7x - 4$ **f** $9 - 5t = 3t - 15$

4 Solve each equation. 9.04

a $2(n + 3) = 18$ **b** $3(x + 1) = 15$ **c** $10(x - 3) = -10$

d $5(x - 2) = 3x + 4$ **e** $4(y + 1) = y + 18$ **f** $-2(d - 2) = 2d - 20$

g $3(r - 1) = 2(r + 9)$ **h** $7(a + 5) = 3(a + 9)$ **i** $2(2n - 4) = 2(5 - n)$

5 Solve each equation: 9.05

a $x^2 = 64$ **b** $x^2 = 45$, correct to one decimal place **c** $5x^2 = 135$, as a surd

6 **a** If a number is doubled and has 4 added to it, the answer is 14. What is the number? 9.06

b If a number has 4 added to it and is doubled, the answer is 14. What is the number?

7 **a** Keilani is paid \$45 for each jumper she knits. If n is the number of jumpers Keilani knits to earn a total of \$270, which equation can be used to find the value of n? Select the correct answer **A**, **B**, **C** or **D**. 9.06

A $\frac{n}{45} = 270$ **B** $45n = 270$ **C** $n + 45 = 270$ **D** $270 - n = 45$

b Solve the equation to find the value of n.

8 7 times a number is the same as 9 more than 4 times the same number. Use an equation to find the number. 9.06

9 The sum (S) of the angles (in degrees) of a polygon is given by $S = 180(n - 2)$, where n is the number of sides. Find: 9.07

a the sum of the interior angles when a polygon has 9 sides

b the number of sides in a polygon whose angle sum is 1080°.

YEAR 9 PATH

10 Change the subject of each formula to the variable in the brackets. 9.08

a $V = IR$ (R) **b** $p = 2q - 5$ (q) **c** $x = 2(y - 3)$ (y)

Foundation Standard Complex

PRACTICE SET 3

ANSWERS ON P. 601

9.02

1 Solve each equation.

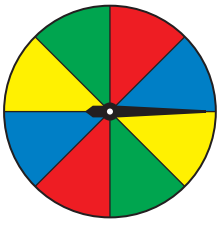
a $2a + 7 = 9$ **b** $3m - 2 = 7$ **c** $3x + 5 = -10$

d $\frac{t}{3} - 5 = 1$ **e** $\frac{k}{7} + 1 = 12$ **f** $8 - 2x = 4$

g $\frac{2x}{3} = 6$ **h** $\frac{5x}{4} = -10$ **i** $\frac{a+3}{2} = 8$

8.01

2 If this spinner is spun, find the probability, as a percentage, that the arrow lands on:



a green

b a traffic light colour

c a colour that is not red

7.01

7.09

3 Mrs Smart gave her class a quick quiz. The results are shown in the table.

a Copy and complete the table.

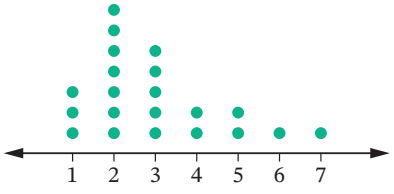
b How many students are there in Mrs Smart's class?

c Construct a frequency histogram and polygon for these results.

Mark	Tally	Frequency
6	𝍸	
7	𝍸 ||	
8		5
9	𝍸 𝍸	
10		3

7.11

4 Describe the shape of the data in this graph. Select the correct answer **A, B, C** or **D**.



A symmetrical

B bimodal

C negatively skewed

D positively skewed

7.01

5 A die was rolled 40 times and the results are shown here.

5	1	5	2	3	2	6	5	2	5
4	2	5	2	6	2	6	4	3	1
5	5	3	6	4	2	6	5	6	5
5	2	3	3	3	3	3	1	2	1

a Draw a frequency table for these results.

b Draw a divided bar graph to represent this information.

Foundation Standard Complex

9780170479516

6 Is each type of data categorical or numerical? 7.03

- **a** number of cars owned
- **b** favourite drink
- **c** type of hairstyle

7 Is each type of data discrete or continuous? 7.03

- **a** lap time of swimmer
- **b** distance from city
- **c** points scored by basketball players

8 A die is rolled. Find the probability that the number that comes up is: 8.01

- **a** even
- **b** at least 2
- **c** a composite number

9 Construct a dot plot for these Mathematics test marks. 7.08

22	16	10	8	12	11	15	13	12	13	11	16
11	12	10	10	12	12	15	11	13	15	17	13

- **a** Find the median.
- **b** Find the mode.
- **c** What is the outlier?
- **d** Where are the marks clustered?

10 Is each type of data as nominal or ordinal? 7.03

- **a** World ranking of a professional tennis player.
- **b** Brand of laptop computer used by Year 8 students.
- **c** Size of pizza.

11 For the data below, find: 7.05

90	89	95	92	90	92	97	93	90	97

- **a** the mean 7.06
- **b** the mode
- **c** the median
- **d** the range

12 A pack of cards contains 10 blue, 6 purple and 4 pink cards. One card is drawn from the pack at random. Find the probability that this card is not purple. 8.02

Foundation Standard Complex

7.07

13 The ages of people taking out a policy with an insurance company are shown.

34	32	28	30	33	28	30	29	33	28	34	34
28	30	32	32	34	31	30	31	28	30	28	30

a Construct a frequency table of the ages, including an 'fx' column.

b What fraction of the people are under 30 years of age?

c For the above data, find:

i the mean (to one decimal place)

ii the median

iii the mode

iv the range.

9.05

14 Solve $3x^2 = 48$. Select the correct answer **A**, **B**, **C** or **D**.

A $x = -2$ **B** $x = 2$ **C** $x = \pm 2$ **D** $x = \pm 4$

9.06

15 5 bricks have the same mass as one brick and 4 kg. What is the mass of one brick? Use an equation to solve this problem.

YEAR 9 PATH

8.04

16 Year 8 students were surveyed on whether they participated in weekend sport.

a Copy and complete this 2-way table.

	Plays weekend sport	Does not play weekend sport	Total
Male	49		
Female		45	84
Total		68	

b How many students are in Year 8?

c How many Year 8 students play weekend sport?

d If a student is chosen at random from Year 8, what is the probability that the student:

i is female?

ii is male or plays sport but not both?

7.12

17 The marks scored by 10 students in Maths and Science exams are shown.

Maths	35	58	87	43	52	96	23	35	48	58
Science	43	69	73	62	54	88	24	40	40	60

a Show these marks on a back-to-back stem-and-leaf plot.

b Find the median and the range for each exam.

c Based on these results, which exam do you think was more difficult? Justify your answer.

18 The nationalities of players in a junior tennis tournament are shown below. 8.05

Australia	USA	UK	Asia	Europe
3	9	4	20	28

Find the probability that a player selected at random from this group:

a is Australian

b comes from Asia

c is not American

d is from Europe or America.

19 Joanna wants to find the most popular TV program in Australia. 7.04

a Should she use a sample or a census? Give a reason for your answer.

b Suppose she decides to use a sample. How could she choose a representative sample?

20 Using the formula for the area of a triangle $A = \frac{1}{2}bh$, find h, the height of the triangle with area 63 and base length 10.5. 9.07

21 Emily is a park ranger. This table shows the data she recorded about kangaroo deaths in the park over one year. 8.06

Cause of death	Number of deaths
Hit by a motor vehicle	78
Shot	12
Caught in a fence or trap	11
Old age	25
Starvation	3
Other	7
No known cause	4
Total	140

a Calculate the observed probability that the death of a kangaroo was caused by a motor vehicle.

b Calculate as a percentage (to the nearest whole number) the observed probability that a kangaroo in the park will die from old age.

c In the next year, there were 90 kangaroo deaths. How many would you expect to have died from being shot?